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Digital twin platform based on an identity-semantic-interaction framework to improve bridge management workflows

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ABSTRACT

Digital technologies have been introduced into infrastructure asset management practices since 1990s. Current digital frameworks are less flexible in large-scale collaborative environments and have limited analytical capability. To address this challenge, this study develops a digital twin platform based on an identity-semantic-interaction framework. It integrates Bridge Information Modeling (BrIM) by applying Building Information Modeling (BIM) principles, along with Internet of Things (IoT) data, and semantic modeling protocols to align multi-source data through a unified property and identifier system, which enables continuous synchronization between the physical bridge and its virtual counterpart. By utilizing capabilities such as parametric modeling, IFC-based interoperability, and cloud-enabled digital-twin coordination, the proposed platform demonstrates a streamlined yet semantically rich digital twin in the bridge engineering domain. The associated workflow monitors in real time and manages inspection records during the entire bridge life cycle and enhances data traceability. Even though limitations remain in scalability, automation, and advanced analytics, the current research significantly lowers the technical barrier to digital twin adoption in bridge projects by eliminating manual IFC data exchange, maintaining stable component identities through PCIDs, and enabling reusable semantic templates that simplify integration of BIM, inspection, and IoT data. It facilitates practical digital twin development and possible future extensions in adaptive modeling, data analytics, and multi-stakeholder collaboration for a proactive asset management.

1. Introduction

Bridges are important in both transportation infrastructure and socioeconomic systems. Nowadays, their deterioration accelerates due to increasing traffic demand, structural aging, lack of maintenance funds, and environmental degradation caused by climate change [1]. Without timely and effective maintenance, such degradation not only increases repair and replacement costs but also poses risks of structural failure, economic loss, and even casualties [2,3]. These challenges call for advanced bridge management systems, which can integrate heterogeneous data and support intelligent decision-making to ensure safety, optimize maintenance strategies, extend the service life of bridges and minimize their life cycle costs [4].

Digital Twin (DT) technology has gained increasing attention as a means to unify sensing data, simulation models, and asset information into a coherent digital representation [5]. The emerging technology integrates design data, inspection records, monitoring information, and simulation results. In this way, it enables management platforms that unite structural health monitoring, risk assessment, maintenance optimization, and performance prediction [6,7]. Despite its evolving definitions, DT is widely considered as "a digital representation of a physical asset that communicates, coordinates, and collaborates within a system to enhance efficiency and knowledge sharing" [8]. In bridge engineering, DT has been increasingly a promising tool in approaching intelligent, data-driven management throughout a structure's lifecycle [9]. When combined with Building Information Modeling (BIM), Internet of Things (IoT), and artificial intelligence technologies, DT supports real-time monitoring, predictive maintenance, and decision support

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[10]. Nevertheless, its practical implementation remains challenging due to issues such as data fragmentation, lack of standardized modeling processes, and limited interoperability among heterogeneous systems [11].

To address the limitations of current digital frameworks in large-scale collaborative environments, this paper evaluates whether an identity-semantic-interaction framework can measurably improve data consistency, interoperability, and synchronization performance as a proof of concept in bridge digital twin workflows compared to conventional BIM-based data exchange methods. The expected advantages include improved identifier persistence, reduced manual data reconciliation, enhanced semantic consistency across platforms, and near real-time synchronization enabled through event-driven integration. To achieve this objective, this study develops a digital twin platform based on an identity-semantic-interaction framework. It integrates the Building Information Modeling (BIM), Internet of Things (IoT) data, and semantic modeling protocols to align multi-source data through a unified property and identifier system. The implementation on an operational bridge (details withheld for confidentiality), verified the practical feasibility of this framework. The framework integrates four key mechanisms; Persistent Component Identifier, Semantic Protocol, Exchange Protocol, and Integration Service, to achieve consistent data continuity across bridge design, inspection, and monitoring stages.

The rest of the study is structured as follows. Section 2 presents a review of existing literature on DT concepts, BIM-based bridge modeling, and the integration of both for bridge lifecycle management. Section 3 illustrates the identity-semantic-interaction framework which includes a seamless data synchronization mechanism. Section 4 describes the case study on an anonymized in-service bridge to demonstrate the functionality of the proposed DT platform, followed by the discussions in Section 5 and conclusions and future work in Section 6.

2. Literature review

2.1. BIM as a foundation for bridge digitalization

Building Information Modeling (BIM) has become a fundamental technology for digitizing the architecture, engineering, and construction (AEC) industry. Through its ability to create object-oriented, semantically rich 3D models, BIM enables multi-dimensional coordination and data-driven decision-making across the design, construction, and maintenance phases [12]. It promotes standardized data exchange and interdisciplinary collaboration among project participants based on Common Data Environment (CDE) systems and the Industry Foundation Classes (IFC) open standard. In bridge engineering, BIM serves to manage geometric, material, structural and operational information during bridge lifecycles [13].

BIM-based bridge projects have shown significant benefits in areas like enhanced visualization, clash detection, scheduling, cost estimation, and construction sequencing [14]. This parametric modeling enables flexible design modifications and scenario simulation, improving efficiency and reducing rework during design and construction [15]. Moreover, 4D and 5D BIM extends traditional geometric modeling to time and cost-based management, improving project planning and progress tracking [16].

However, as transitioning into the operation and maintenance (O&M) stage, BIM's potential remains underused [17]. Although large quantities of inspection and monitoring data are collected during service life, such information is often fragmented, static, and non-interoperable [18]. Current BIM applications in O&M mainly function as visualization or documentation repositories. They lack real-time interaction with structural health data or environmental monitoring systems [19]. And these limitations constrain BIM from supporting maintenance and making adaptive decisions. Consequently, the AEC community has turned to Digital Twin (DT) technology to overcome these shortcomings and realize a continuous digital representation of bridge assets [20].

2.2. Evolution of BIM–DT integration in bridge management

DT extends the concept of BIM by supporting continuous synchronization between the physical and virtual environments [21]. Unlike static BIM models, DT models continuously, synchronize with real-world conditions through sensor networks, Internet of Things (IoT) devices, and artificial intelligence (AI) algorithms [22]. This helps transform passive storage data into active operational intelligence, thus supporting real-time monitoring, performance simulation, and intelligent feedback mechanisms [23].

In bridge management, several pioneering studies have explored the integration of BIM and DT paradigms. A hybrid BIM–DT resource management model is proposed which dynamically reallocates resources using AI-assisted automation and self-updating mechanisms [24]. The integration of IoT and AI is identified in the literature as a critical driver for the semantic transition from BIM to Digital Twins, enabling a dynamic and intelligent flow of information [25]. Apart from this, a multi-source data fusion framework has been established to achieve comprehensive lifecycle analysis by unifying BIM models, sensor streams, and simulation results within one single digital twin [26]. These efforts collectively mark a paradigm shift, from static digital modeling to dynamic digital representation, where information evolves alongside the physical structure.

Digital twin maturity models have been proposed to classify the levels of sophistication in implementation. According to Tao et al. [27], the maturity hierarchy consists of four stages: pre-digital twin, digital twin, adaptive twin, and intelligent twin. Most bridge applications currently lie between the first and second stages, characterized by periodically updated BIM-based models with limited automation and analytical capabilities [28]. Advanced "adaptive twins" incorporate near real-time sensor feedback and predictive analytics. "Intelligent twins" integrate AI-driven reasoning and autonomous decision-making [29]. Progressing toward the higher maturity levels demands not only technological integration but also standardized data structures, semantic consistency, and interoperable communication protocols. Currently these problems remain insufficiently addressed.

2.3. Data integration, interoperability, and model synchronization

A major barrier to implementing bridge digital twins lies in the heterogeneity of data sources. Bridge management involves diverse datasets, including as-designed BIM models, as-built records, inspection photos, finite element (FE) simulations, and sensor-based monitoring data [30]. Integrating these heterogeneous sources requires robust data fusion strategies and standardized interfaces. Pashoutani et al. [31] developed FE-based digital representations by updating structural models using field measurements, but these models are often limited by software dependency and lack of interoperability across platforms. A point-cloud-based modeling method has been developed to automatically reconstruct bridge geometry for digital twin applications [32]; however, this approach still lacks mechanisms for semantic enrichment and continuous synchronization. Likewise, surrogate modeling and machine learning techniques have been utilized to build FEM–DT systems for maintenance prediction [33], but these methods remain domain-specific and disconnected from BIM-based asset information.

These fragmented approaches show that there is still no clear and consistent process for building bridge digital twins. Data often exist in silos, maintained by different tools and stakeholders, posing challenges to achieving consistent data exchange and version control throughout the lifecycle. Moreover, the lack of persistent identity systems, where physical and digital components share a unique identifier, makes it difficult to track updates and associate monitoring data with corresponding structural elements.

2.4. Semantic modeling and interoperability standards

Semantic interoperability represents another key research frontier in DT development. The Industry Foundation Classes (IFC) standard, as an open and neutral data model, serves as the primary format for data exchange in BIM-based systems [34]. IFC provides an ontology for representing geometric and non-geometric attributes of infrastructure elements, supporting consistent data sharing across different software environments. However, IFC's current schema was primarily developed for design and construction phases. It exhibits limitations in dynamic updates, time-series data integration, and scenario simulation, all of which are essential capabilities for real-time digital twins [35].

Several studies have attempted to extend IFC for infrastructure applications. For example, the IFC Bridge extension under the IFC 4.3 schema introduces new entity classes to support bridge-specific elements, while other researchers have proposed semantic mapping methods to integrate IFC with Internet of Things (IoT) data streams [36, 37]. Despite these efforts, challenges remain in aligning real-time monitoring data with the static BIM structure, particularly when managing temporal changes and multi-resolution data from diverse sources.

Semantic modeling frameworks, such as ontologies, knowledge graphs, and standardized property sets, are increasingly being adopted to enhance the machine-readability and semantic richness of digital twins [38]. These approaches facilitate cross-domain data fusion and enable higher-level reasoning for maintenance planning, anomaly detection, and lifecycle optimization [39]. Nonetheless, the lack of a consensus semantic protocol for bridges continues to impede interoperability among BIM, IoT, and analytical systems.

2.5. Challenges in current bridge digital twin development

Despite these advancements, several challenges hinder the practical implementation of bridge digital twins. A key limitation is the lack of a unified framework for data exchange and interoperability. For instance, Lin et al. [40] constructed finite element (FE) models based on design documentation and updated them with measured data, using these as DT representations. Although valuable for performance analysis, such models often suffer from software dependency and weak interoperability. Ma et al. [41] developed automated model-fitting methods based on point cloud data and identified these models as digital twins, but they still lacked mechanisms for dynamic data fusion and real-time synchronization. Yang et al. [42] introduced FEM–DT systems using surrogate models for maintenance prediction, yet these approaches remain fragmented, relying on diverse modeling and communication methods.

This fragmentation reflects deeper issues: the absence of standardized workflows, persistent identification systems, unified data structures, and semantic communication protocols. Without these, data interoperability and cross-platform collaboration remain severely constrained. And the continuity of information throughout the bridge lifecycle will be disrupted.

From a standardization perspective, the IFC provides a globally recognized schema for data interoperability in construction and asset management [43]. While IFC lays a solid foundation for digital twin development, its current schema does not fully support dynamic updates, real-time synchronization, and hypothetical scenario simulation [44]. Moreover, the fast-changing software environment and cross-domain integration further complicate interoperability and long-term knowledge preservation [45].

Based on the preceding research, the identified gap lies in the absence of a standardized and semantically enriched framework integrating BIM-based static information with DT-enabled dynamic data. Such a framework is essential to ensure persistent identity, semantic interoperability, and real-time interaction throughout all stages of the bridge lifecycle. Addressing this gap constitutes the core objective of the present study. The novelty of this study lies not merely in combining BIM and digital twin concepts, but in proposing a standardized and semantically enriched framework that enables the integration of BIM-based static information with digital twin-enabled dynamic data across the bridge lifecycle. Specifically, the framework formalizes the integration of identity persistence (through unique and stable component identifiers), semantic invariance (ensuring consistent meaning of data across platforms), and event-driven protocols (enabling real-time system interaction and responsiveness) within a single unified architecture. This integrated approach addresses key limitations in existing studies, which are often fragmented in terms of modeling strategies, data exchange methods, and interoperability mechanisms.

3. Methodology

Building upon the above analysis, this study aims to address the limitations of existing BIM environments by proposing a unified framework for BIM-DT integration. The framework incorporates consistent identifiers, semantic alignment, and standardized data exchange protocols to bridge the gap between static BIM models and dynamic digital twin systems. To be specific, the proposed method consists of four main components:

3.1. Persistent component ID

Design models in 3D modeling software like Revit or BlenderBIM often use transient element IDs. These IDs are regenerated when the model is recompiled, causing identity loss once data are exported to other systems such as IFC or Tandem. Similarly, inspection and monitoring data are usually recorded in spreadsheets or databases without any formal link to the geometric instance that they describe. To solve this issue, a persistent component ID (PCID) is designed to maintain the identity of each component as data migrate between platforms and phases. Each bridge component, such as a girder, pier, or deck slab, is assigned a PCID with the following encoding rule:

$$PCID = GBS_ID + PBS_ID + ComponentCode + UniqueID$$

where 'GBS_ID' denotes the geometric subsystem, 'PBS_ID' represents the product breakdown structure, and the 'ComponentCode' and 'UniqueID' specify element type and sequence. For example, the identifier PEI-PC-RT12-BR_051-SP1-STR-SP-GIR-NBT_1000-PC-001 refers to a precast (PC) NBT-1000 girder (GIR) #001 within the superstructure (SP), functioning as a structural (STR) component in span #1 (SP1) of the bridge registered as 51 (BR_051), and geographically located on Route 12 (RT12) within a specific county (PC) and province (PEI) in the country.

The PCID functions as a cross-domain identifier that remains stable throughout all transformations and updates. By embedding this identifier as a shared parameter in the model and recording it within the IFC property set, the PCID remains consistent across downstream databases and Application Programming Interfaces (APIs). In this way, PCID serves as a digital anchor for data traceability in the bridge digital twin. Inspection records, sensor data, and simulation results can be linked to the same component. This mechanism enables continuous data synchronization to keep the physical and digital domains consistent.

The proposed PCID coding structure is the result of engineering practice and theoretical alignment. First, its architectural design is highly compatible with established asset management hierarchy practices; the GBS_ID (Geometric Subsystem) directly maps to the spatial decomposition logic of the IFC 4.3 standard (e.g., *IfcBridge* → *IfcBridgePart* → *IfcElement*). Simultaneously, the PBS_ID (Product Breakdown Structure) ensures logical interoperability with bridge maintenance databases, ensuring native consistency of the semantic model within the industry-standard framework. Second, the PCID is designed to address the 'transience' issue inherent in traditional BIM element identifiers (such as GUIDs) during cross-platform interaction and model regeneration. As standard BIM IDs are susceptible to change

due to software exports or version updates, which can break the physical link between geometric components and monitoring data. The PCID employs a 'ComponentCode + UniqueID' combination to provide a persistent digital anchor independent of the underlying software. This design balances rigor with operational practicality. Compared to relying on complex ontology-based reasoning, this approach satisfies the structural requirements of IFC entity decomposition while significantly lowering the maintenance threshold for practitioners in real-world bridge management scenarios. Ultimately, by embedding the PCID into IFC property sets, this study enables efficient cross-lifecycle data traceability and continuous synchronization between physical and virtual spaces, providing a robust logical foundation for the implementation of the proposed digital twin framework.

3.2. Semantic protocol based on IFC 4.3

To realize the interoperation ability, a semantic protocol is needed. The semantic protocol is defined as a structured interoperability mechanism. It formalizes the meaning, structure, and constraints of data properties across heterogeneous systems through schema-based mappings. It also ensures that identical engineering concepts such as *material grade, inspection date*, or *defect severity*, retain consistent semantics and relational context whether stored within a BIM environment, exchanged via an IFC file, or queried from a database.

In this paper, the semantic protocol operationalizes this consistency through JSON-based mapping templates aligned with the IFC 4.3 schema. Each template specifies a property's name, data type, unit, constraint rule, and relational context (e.g., component hierarchy or inspection record linkage). For example, a typical mapping template may define how a model parameter such as BridgeSpanLength corresponds to the IFC attribute IfcBeam.Length, with its data type set as float and unit as meter, or how a field like InspectionDate links to an inspection record associated with a specific component ID. The template thereby serves as a semantic contract that governs how data are interpreted and exchanged between modeling tools (BlenderBIM), neutral models (IFC), and analytical environments (Power BI, Tandem). Mathematically, the semantic protocol can be expressed as a mapping function:

$$S = \{P_i\}_{BIM} \rightarrow \{P_j\}_{IFC} \rightarrow \{P_k\}_{DB}$$

where each property P maintains invariant semantics under transformation, ensuring that semantic equivalence holds across all representational domains. Unlike traditional ontology-based models (e.g., IfcOWL, BOT), the semantic protocol adopts a *template-driven* approach emphasizing operational interoperability rather than exhaustive semantic formalization. The definition of the semantic protocol is shown in Table 1.

Fig. 1 illustrates an example of how the template maps a bridge girder component identified by ComponentID across multiple systems. The ComponentID from the model is automatically translated to ComponentID_IFC in the IFC schema and ComponentID_Tandem in the Tandem digital twin platform. Additional properties, such as MaterialGrade and InspectionDate are similarly mapped with proper data type, unit, and transformation rules. The semantic protocol also incorporates validation rules and evaluation metrics to verify identifier persistence, property mapping accuracy, and synchronization performance during integration workflows. This cross-platform mapping ensures that all systems consistently refer to the same physical component, enabling synchronized updates, semantic consistency, and progressive information addition throughout the bridge lifecycle.

3.3. Integrated data exchange protocol

In much of the research on bridge digital twins, IFC is mentioned and used to implement data exchange. Though IFC provides a standardized, schema-based structure for data encoding, it remains a file-centric and batch-oriented mechanism. It is not sufficient for the integration because its workflows rely on manual export/import steps, thus introducing latency and version inconsistencies. In a digital twin context, the data need to flow seamlessly across design, monitoring, and analytics environments. Therefore, in this research, a data exchange protocol is proposed which defines the data flow through a standardized message structure:

$$EP = \{Format,\ Method,\ Endpoint,\ Trigger,\ Response\}$$

where data format (e.g., JSON, CSV, MQTT), transmission method (e.g., HTTP POST), and event triggers (e.g., event warning, scheduled update) are included. The detailed definition of EP is as follows in Table 2:

Using the example in Fig. 2, this data exchange template follows the six elements defined in the table: Format is JSON, indicating that the data is transmitted in a structured key-value format; Method is HTTP POST, used to send sensor data to the target system; Endpoint points to

Table 1
Definition of fields in semantic mapping template.

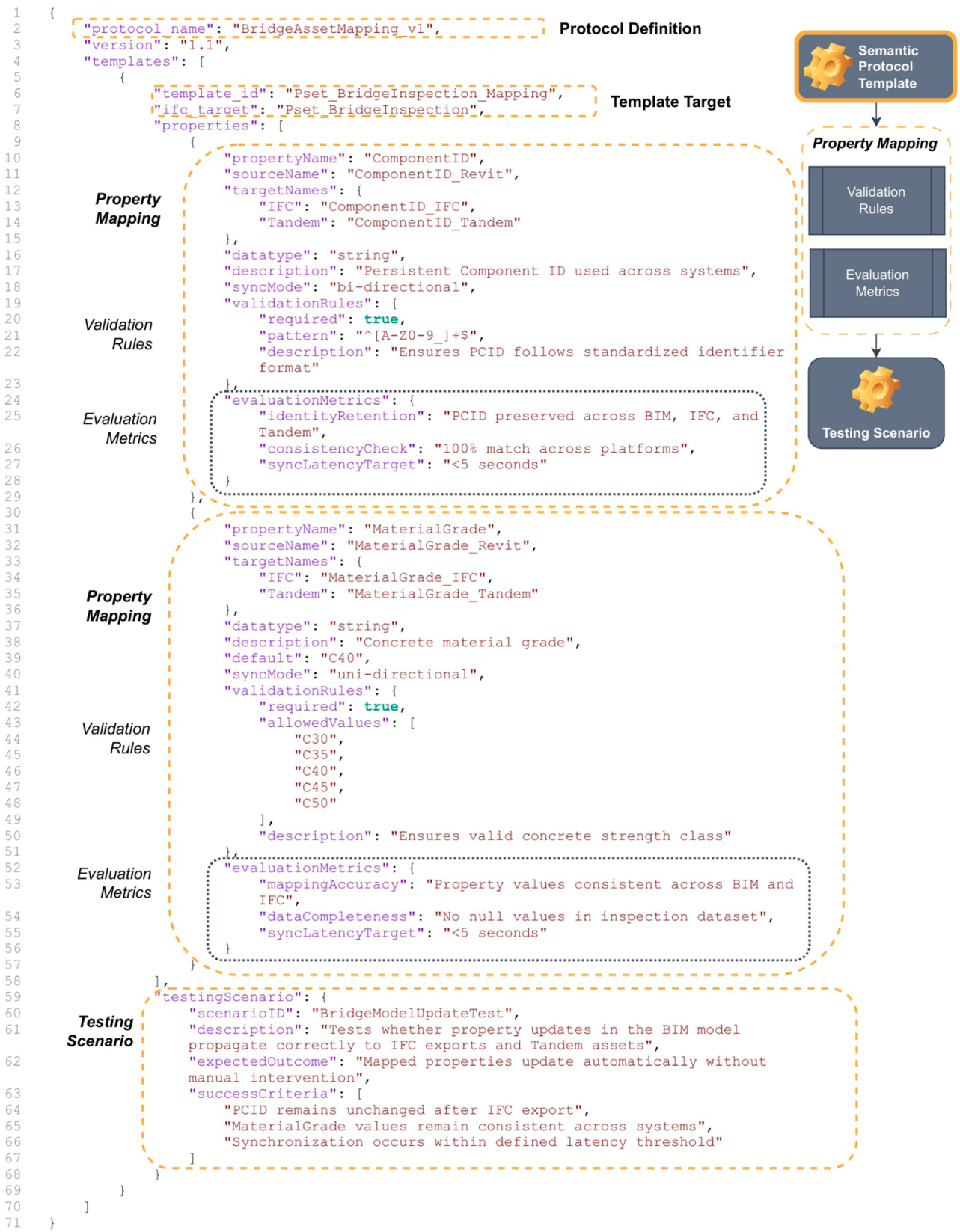
Field Name	Data Type	Description / Example
protocol_name	string	Name of the semantic protocol. Example: "BridgeAssetMapping_v1"
version	string	Version identifier of the protocol. Example: "1.0″
description	string	Short description of the protocol. Example: "Mapping templates aligned with IFC 4.3 for bridge inspection"
templates	array	A list of mapping templates, each defining property correspondence between BIM and IFC.

Table 1(a). Template-Level Fields

Field Name	Data Type	Description / Example
template_id	string	Unique identifier of the template. Example: "Pset_BridgeInspection_Mapping"
ifc_target	string	Target IFC Property Set for mapping. Example: "Pset_BridgeInspection"
properties	array	List of property mappings defined within the template.
testingScenario	object	Defines the evaluation scenario used to verify mapping behavior.

Table 1(b). Property-Level Fields

Field Name	Data Type	Description / Example
propertyName	string	Target property name in IFC or semantic model. Example: "CrackWidth_mm"
sourceName	string	Corresponding field name in the source data. Example: "CrackWidth"
dataType	string	Data type of the property (e.g., float, string, integer). Example: "float"
unit	string	Unit of measurement or data format. Example: "mm", "MPa", "YYYY-MM-DD"
constraint	object	Rules defining allowed range or enumeration. Example: {"min": 0, "max": 100}
relation	object	Semantic relationship or hierarchy context. Example: {"componentHierarchy": "Superstructure/Girder"}
description	string	Explanation of the property's engineering meaning. Example: "Measured crack width in millimeters"
syncMode	string	Direction of data synchronization. Example: "bi-directional"
updatePolicy	string	Defines how updates are handled. Example: "append_history"
default	any	Default value when source data is missing. Example: "C40"
transform	string	Transformation function for data conversion. Example: "toFloat"
validationRules	object	Defines validation conditions such as required fields, value ranges, or pattern constraints. Example: "allowedValues"
evaluationMetrics	object	Defines performance indicators used to assess mapping performance




```
{
  "protocol_name": "BridgeAssetMapping_v1",
  "version": "1.1",
  "templates": [
    {
      "template_id": "Pset_BridgeInspection_Mapping",
      "ifc_target": "Pset_BridgeInspection",
      "properties": [
        {
          "propertyName": "ComponentID",
          "sourceName": "ComponentID_Revit",
          "targetNames": {
            "IFC": "ComponentID_IFC",
            "Tandem": "ComponentID_Tandem"
          },
          "datatype": "string",
          "description": "Persistent Component ID used across systems",
          "syncMode": "bi-directional",
          "validationRules": {
            "required": true,
            "pattern": "^[A-Z0-9_]+$",
            "description": "Ensures PCID follows standardized identifier format"
          },
          "evaluationMetrics": {
            "identityRetention": "PCID preserved across BIM, IFC, and Tandem",
            "consistencyCheck": "100% match across platforms",
            "syncLatencyTarget": "<5 seconds"
          }
        },
        {
          "propertyName": "MaterialGrade",
          "sourceName": "MaterialGrade_Revit",
          "targetNames": {
            "IFC": "MaterialGrade_IFC",
            "Tandem": "MaterialGrade_Tandem"
          },
          "datatype": "string",
          "description": "Concrete material grade",
          "default": "C40",
          "syncMode": "uni-directional",
          "validationRules": {
            "required": true,
            "allowedValues": [
              "C30",
              "C35",
              "C40",
              "C45",
              "C50"
            ],
            "description": "Ensures valid concrete strength class"
          },
          "evaluationMetrics": {
            "mappingAccuracy": "Property values consistent across BIM and IFC",
            "dataCompleteness": "No null values in inspection dataset",
            "syncLatencyTarget": "<5 seconds"
          }
        }
      ],
      "testingScenario": {
        "scenarioID": "BridgeModelUpdateTest",
        "description": "Tests whether property updates in the BIM model propagate correctly to IFC exports and Tandem assets",
        "expectedOutcome": "Mapped properties update automatically without manual intervention",
        "successCriteria": [
          "PCID remains unchanged after IFC export",
          "MaterialGrade values remain consistent across systems",
          "Synchronization occurs within defined latency threshold"
        ]
      }
    }
  ]
}
```

Fig. 1. Example of semantic protocol.

Tandem's asset update API:

(https://api.tandem.autodesk.com/assets/update); Trigger is OnSensorThreshold, which initiates data transmission when the sensor on a component exceeds a predefined threshold; Payload contains the actual data {"PCID": "PEI-PC-RT12-BR_051-SP1-STR-SP-GIR-NBT_1000-PC-001″, "strain": 45.7}, indicating that the strain of girder PEI-PC-RT12-SP1-STR-SP-GIR-NBT_1000-PC-001 is 45.7 με; and Response returns system feedback {"status": "success", "GUID": "A2C4…"}, confirming that

Table 2
Definition and application of the exchange protocol (EP) template.

Element	Definition	Example (Based on Fig. 2)
Format	Specifies the data encoding or message format used for transmission. Can be structured (e.g., JSON, CSV) or stream-based (e.g., Message Queuing Telemetry Transport (MQTT)).	JSON for API payloads to Tandem; MQTT message for IoT stream; .xlsx for offline inspection upload.
Method	Defines the communication mechanism or protocol used to send or receive data.	HTTP POST for sending sensor data updates to Tandem API; GET for fetching BIM element changes via the add-on in BlenderBIM.
Endpoint	The URL, API path, or topic name that uniquely identifies the data target or destination.	https://api.tandem.autodesk.com/assets/update (Tandem API); /bridge/PEI-PC-RT12-BR_051/girder/strain (MQTT topic); /datasets/{id}/rows (Power BI dataset).
Trigger	Defines the event condition that initiates data exchange. It can be periodic, event-based, or threshold-based.	OnFileUpdate (CDE IFC update event); OnSensorThreshold (strain > 50 με); OnDesignChange (model revision).
Timestamp	The date and time when the event message is generated or transmitted	"timestamp": "2025–12–24T10:42:15Z"
Payload	The actual data transmitted to the target system, including identifiers, sensor readings, or attribute values.	{"PCID": "PEI-PC-RT12-BR_051-SP1-STR-SP-GIR-NBT_1000-PC-001″, "strain": 45.7}
Response	Describes the expected system feedback after data transmission.	Tandem API returns {"status": "success", "GUID": "A2C4…"}; Power BI returns dataset refresh confirmation.

the data has been successfully updated, with GUID being the global unique identifier for the processed update. In this manner, the integration service enables seamless synchronization of IoT data with the digital twin model along the "geometry–attribute–behavior–decision" continuum, ensuring both data traceability and consistency.

3.4. Seamless data synchronization mechanism with integration service

An Integration Service is proposed in this paper, which functions as the middleware that implements information synchronization across heterogeneous platforms within the bridge digital twin ecosystem. In this digital twin ecosystem, BlenderBIM, Autodesk Docs (CDE), Tandem, Power BI, and IoT data streams are utilized, and the integration service is employed to enable both *file-based* and *API-driven* interoperability without manual manipulation.

As illustrated in Fig. 3, the service operates with a six-stage pipeline, ensuring data consistency and traceability throughout the "geometry–attribute–behavior–decision" continuum. The six-stage pipeline is described as follows:

Step 1: Event Detection

The workflow begins with the detection of data or model change events. These events originate from multiple sources: (1) model revision events from BlenderBIM via its Add-on(e.g., element modification or IFC re-export); (2) file versioning events captured by the CDE (e.g., updates to IFC or BIM Collaboration Format (BCF) files); and (3) sensor or inspection updates from IoT databases or field reports. Each detected event triggers the corresponding Exchange Protocol (EP) configuration, specifying the target endpoint and synchronization rule.

Step 2: Data Snapshot

Once an event is detected, the Integration Service captures a data snapshot, which is a structured representation of the changed object or record. For design and model updates, this snapshot may include model component geometry, PCID, and associated metadata extracted via BlenderBIM add-on or IFC file parsing. For IoT and inspection updates, it captures sensor readings, timestamps, and inspection attributes such as defect type and severity. All snapshots are normalized and stored in a temporary database table to enable semantic comparison and rollback tracking.

Step 3: Semantic Mapping

At this stage, the Integration Service performs schema alignment using the Semantic Protocol. Each property (e.g., "MaterialGrade", "InspectionDate", "CrackWidth_mm") is mapped to the corresponding IFC 4.3 property set or BCF tag according to the JSON-based mapping templates. This ensures that all attributes conform to a unified schema across IFC, Tandem, and Power BI environments. For example, a

```
{
"Format": "JSON",
"Method": "POST",
"Endpoint": "https://api.tandem.autodesk.com/assets/update",
"Trigger": "OnSensorThreshold",

"Payload": {
"PCID": "PEI-PC-RT12-BR_051-SP1-STR-SP-GIR-NBT_1000-PC-001",
"strain": 45.7,
"timestamp": "2025-12-26T10:42:15Z"
},

"validationStatus": "passed",

"Action": "UpdateAssetProperty",

"Response": {
"status": "success",
"GUID": "A2C4..."
}
}
```

Fig. 2. Example of exchange protocol.

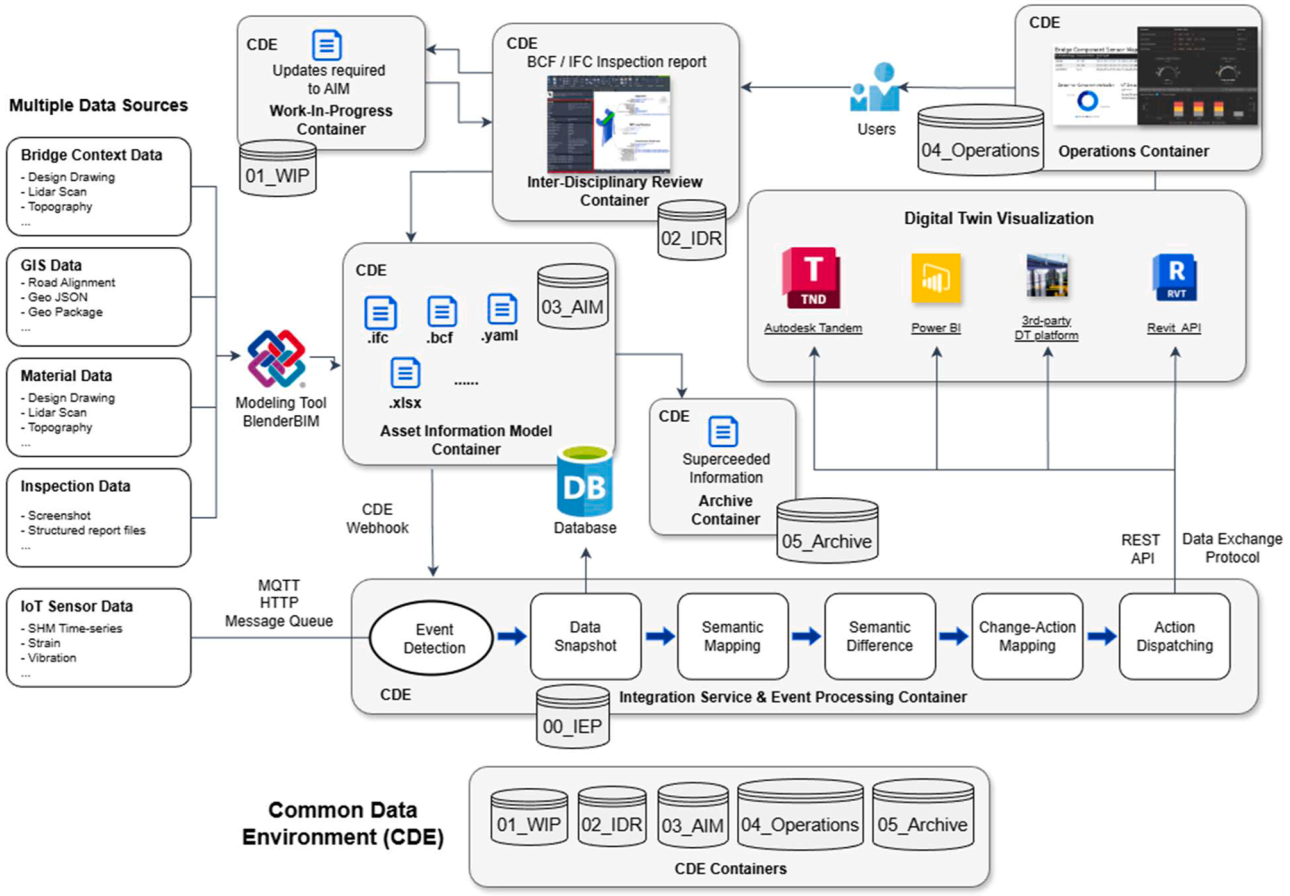


Fig. 3. Integration service pipeline for bridge digital twin ecosystem.

parameter in a field inspection form can be semantically mapped to Pset_BridgeInspection in IFC or to a property field in Tandem. Each semantic template includes TemplateVersion, IFCVersion (e.g., IFC 4.3.2.0), RevisionDate, and ChangeLog in alignment with ISO 19,650 Information Management best practices. Templates are stored in a version-controlled repository (e.g., Git-based system). This allows tracking modifications and rolling back to earlier versions if desired.

For handling the IFC schema updates, if a new IFC schema version is released a new template version is created and only affected mappings are updated so backward compatibility with previous IFC exports can still occur and thus no disruption to existing digital twin deployments. When handling Project-Specific Extensions, if a project introduces new properties (e.g., fatigue category for steel bridges), a project-specific template extension is created, the base template remains unchanged and the extension inherits the same mapping structure so there is adaptation to different bridge types and scalability without redesigning the entire semantic protocol.

Step 4: Semantic Difference

The Semantic Difference module compares the latest snapshot with the previous version of the same PCID instance to detect meaningful changes. This process distinguishes *semantic deltas*, such as property value updates, new defect entries, or geometric alterations, from non-impacting metadata changes. For design workflows, it can identify when a girder's material or geometry has been revised in modeling tool and trigger an IFC regeneration. For maintenance workflows, it can detect when an inspection file adds a new defect record and prepares a BCF issue creation.

Step 5: Change–Action Mapping

Each detected semantic delta is then associated with an appropriate action type according to pre-defined rules in the Exchange Protocol configuration. For example, a model geometry change will trigger IFC Export and CDE upload; an IFC version change will trigger Tandem model refresh; an inspection report with a warning (condition state change or intervention requirement) will trigger BCF issue creation and Power BI dataset update; a sensor threshold exceedance will trigger Alert dispatch to Tandem and Power BI. This mapping turns system-level differences into workflow-specific operations, enabling coordinated updates across the digital twin ecosystem.

Step 6: Action Dispatching

In the final step, the Integration Service executes the prescribed actions using API calls or file operations. For example, invoke Tandem API for real-time asset updates, Power BI Representational State Transfer (REST) endpoint for analytics refresh, and BlenderBIM add-on for model synchronization or version tagging. Each action response is logged for verification and rollback if required.

Through this mechanism, the Integration Service maintains continuous, event-driven synchronization across design, monitoring, and decision layers. It transforms the bridge digital twin into a data ecosystem with real-time adaptability, semantic consistency, and lifecycle traceability. To ensure framework versatility and handle the evolution of semantic templates, the system incorporates a modular management mechanism. This mechanism allows the framework to adapt when project requirements change. For example, when replacing a concrete girder bridge with a steel truss bridge, the system simply remaps component classes to corresponding IFC entities, such as IfcMember for truss members, IfcPlate for gusset plates, and IfcFastener for bolted connections. Most importantly, the PCID encoding structure remains consistent throughout these transitions. By maintaining a uniform format of 'GBS_ID', 'PBS_ID', 'ComponentType', and 'UniqueID', the framework ensures identification continuity across the system

hierarchy. Adaptability at the template level is achieved through extensible property sets, which accommodate specific material and structural parameters like fatigue categories, weld types, bolt grades, and corrosion coating classes. Consequently, the digital twin platform preserves semantic consistency and architectural flexibility, even in diverse and complex bridge engineering scenarios.

3.5. Validation methodology

To evaluate the effectiveness of the proposed framework, a proof-of-concept validation methodology was adopted, combining controlled system testing with real-world implementation on an operational bridge. The validation focuses on assessing whether the framework improves data consistency, interoperability, and synchronization efficiency compared to conventional BIM-based workflows.

The validation was conducted through a series of controlled integration scenarios, including BIM model updates, sensor data ingestion, IFC-based inspection workflows, and BCF issue coordination. These scenarios simulate typical lifecycle events in bridge management and enable systematic observation of data propagation across platforms. The evaluation criteria used to evaluate the framework can be seen in Table 3.

Validation results were derived from system logs, automated rule checks embedded in the semantic protocol and observed workflow execution within the case study environment. The implementation was internally validated within the project team, and is intended as a foundation for future external validation through engagement with infrastructure owners and domain experts.

While Section 3 defines the proposed identity–semantic–interaction framework and its implementation mechanisms, it does not in itself demonstrate performance or effectiveness. To address this, the following section presents a structured validation through a proof-of-concept case study, where the framework is applied and evaluated against defined performance criteria, including identifier persistence, semantic consistency, synchronization latency, and data completeness.

4. Case study: example bridge digital twin

The practical applicability of the proposed identity–semantic–interaction framework and standardized workflow was evaluated through a case study conducted on a representative three-span reinforced concrete highway bridge with precast prestressed concrete girders and cast-in-place reinforced concrete decks. While the previous section described the system architecture and integration approach, this section focuses on evaluating its performance under real-world and controlled scenarios, acting as a proof-of-concept. The case study is designed to assess whether the framework improves data consistency, interoperability, and synchronization efficiency compared to conventional BIM-based workflows. The structure includes 1067 individual components across superstructure and substructure elements. Based on design documentation in PDF formats, a fully parametric 3D BIM model was developed as is shown in Fig. 4. A sensor network was deployed, consisting of eight types of sensors—temperature, strain, displacement, corrosion potential, acceleration, humidity, and vibration—distributed across 112 monitored components. The integration service proposed by this paper is developed by Python 3.12, and the service is hosted on Google Cloud Platform. The database used in this case study is Cloud SQL PostgreSQL. The sensor data is stored in BigQuery. The case study demonstrates how the theoretical mechanisms and workflow layers introduced earlier were implemented to test the feasibility of real-world full-lifecycle digital twin integration using open BIM standards and IoT data.

Table 3
Validation framework and evaluation criteria.

Evaluation Criteria	Description	Evaluation Method
Identifier Persistence	Ability to maintain stable PCIDs across BIM, IFC, and DT platforms	Track component IDs during model updates, IFC exports, and data synchronization
Semantic Consistency	Consistency of property meaning and values across platforms	Compare mapped attributes between BIM model, IFC schema, and DT platform
Synchronization Latency	Time required for data updates to propagate across systems	Measure time between event trigger (e.g., sensor update) and system response
Data Completeness	Integrity of transferred data without missing values	Validate datasets after synchronization using rule-based checks
Automation Efficiency	Reduction in manual intervention compared to traditional workflows	Compare number of manual steps vs automated pipeline execution
System Robustness	Ability to handle repeated updates and cross-platform interactions	Perform randomized update tests (e.g., IFC regeneration, sensor triggers)

To ensure a systematic evaluation, the case study is structured around predefined validation scenarios and performance criteria. These criteria are used consistently across all workflows to assess the effectiveness of the proposed framework.

4.1. Application in four case scenarios

Based on the architecture mentioned above, the data synchronization across the BIM model design, status monitoring, model inspection and collaboration is automated. In this paper, four representative application scenarios are mainly considered that demonstrate how this architecture operates under different types of change events: (1) BIM model revision, (2) sensor data update, (3) IFC inspection workflow, and (4) BCF coordination workflow.

4.1.1. BIM model revision

As illustrated in Fig. 5, when a designer updates a component in the 3D modeling tool—BlenderBIM in this case study—such as modifying material grade, geometry, or reinforcement details, the Integration Service detects the modification through a local event-capture mechanism in the add-on, which is similar to the Revit API event listener in Revit workflows. Each modified element carries a Persistent Component ID (PCID), allowing the system to capture a structured data snapshot containing its parameters and metadata. This snapshot is semantically aligned with the IFC 4.3 schema using the Semantic Protocol, ensuring that fields such as “MaterialGrade” and “Volume” retain consistent meaning across platforms. The service then performs semantic difference analysis, comparing the new snapshot with the previous version to identify actual engineering changes rather than cosmetic ones. Once confirmed, the Change–Action Mapping module triggers an automatic IFC export for the updated model. The resulting IFC file is uploaded to the Common Data Environment (CDE), where a webhook notifies the Integration Service of the new version. The add-on installed on another user’s computer can also receive notifications from the Integration Service and automatically retrieve the updated IFC file from the CDE, enabling seamless team coordination. Finally, in this project, the online digital twin platform used for demonstration is Tandem. The Integration Service synchronizes the modified IFC file to Autodesk Docs, from which Tandem automatically retrieves the updated model to refresh the digital twin, ensuring alignment between the BIM design model and its digital twin representation. Autodesk Docs is used here to trigger model updates in Tandem as part of the demonstration workflow. If time permits, an independent online digital twin platform could be developed in the future, seamlessly integrating with the Integration Service's message-push mechanism.

4.1.2. Sensor data update

For real-time monitoring, bridge components are linked to sensors through their PCIDs. When a new measurement, such as strain or temperature, is published by the IoT system, the Integration Service detects

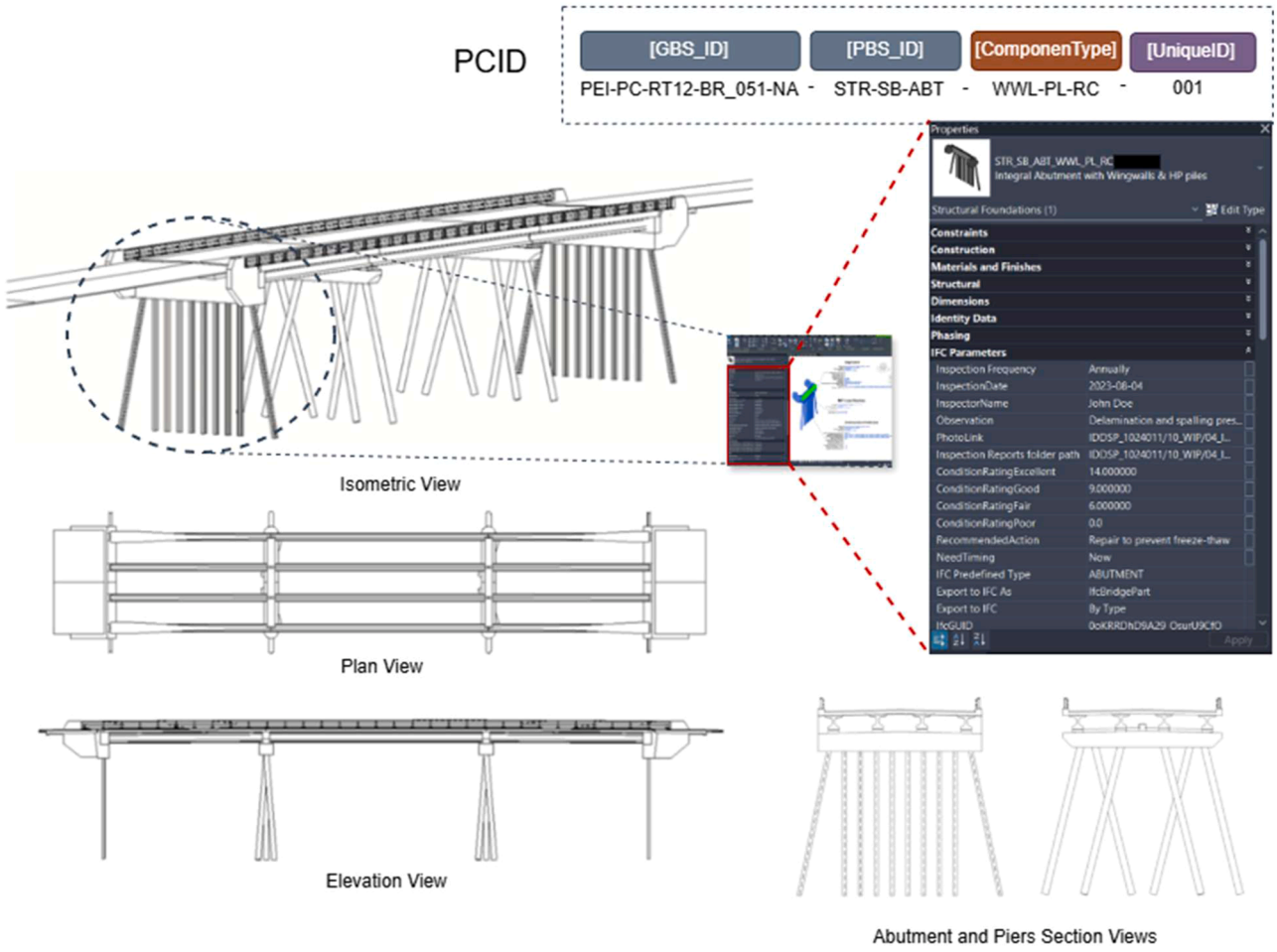


Fig. 4. Views of the Example Bridge from Different Perspectives.

the event through the Exchange Protocol defined trigger OnSensorUpdate. As shown in Fig. 6, the service captures a data snapshot containing the sensor value and timestamp. It then performs semantic mapping to associate it with the corresponding PCID-defined component. If the sensor value exceeds a defined threshold, the Semantic Difference and Change–Action Mapping modules classify the update as a critical event. An alert message is then dispatched to both Tandem via API update of asset properties and Power BI via Representational State Transfer (REST) push dataset, allowing real-time visualization and alarm triggering on both platforms. This workflow enables the digital twin to continuously reflect the bridge's live condition, providing instant feedback for decision-making and maintenance planning.

4.1.3. *IFC inspection workflow*

As shown in Fig. 7, field engineers first transform their spreadsheet-based inspection reports into IFC files using the IfcOpenShell-based Python script developed in this study and subsequently upload the IFC inspection results to the CDE. (e.g., InspectionReport.ifc). The Integration Service continuously monitors the CDE repository through webhook notifications. When a new IFC file is detected, users on other machines can automatically load it into their modeling tools through the add-on. At the same time, the service captures a data snapshot by parsing its property sets, such as Pset_BridgeInspection, and extracting relevant attributes (inspection date, defect type, severity, and notes). Using the Semantic Protocol, these attributes are mapped to the standardized property schema in Tandem. The Semantic Difference module compares the new inspection file with the previous version to identify added or updated defect records. Detected changes trigger the Change–Action Mapping stage, which initiates a Tandem API update to synchronize the inspection results with the corresponding bridge components. In cases where a defect requires follow-up action, a BCF issue can also be automatically generated and linked to the same PCID, enabling cross-platform coordination between inspectors and designers. BCF issue coordination adopts a semi-automated workflow to balance efficiency with engineering rigor. The system filters and prioritizes events based on pre-set thresholds and sensor-triggered rules. To maintain accountability, experts must review and approve these automatically generated issues before they are finalized by integrating a RACI matrix – a framework that defines who is Responsible for carrying out the work, Accountable for final approval, Consulted for expert input, and Informed of outcomes, and is supported by and a clear distribution list. This hybrid approach ensures that critical defects are accurately assigned to the responsible stakeholders while preventing oversight.

4.1.4. *BCF coordination workflow*

The BCF workflow (shown in Fig. 8) complements the IFC inspection process by managing issue tracking and collaboration. When a new issue is created in Tandem or Power BI, such as "surface crack detected on Girder #001″, the Integration Service detects the event through the BCF API. The issue metadata, including PCID, description, and status, is semantically aligned with IFC attributes to maintain consistency between inspection and design layers. If the issue requires a model correction, the engineer on another workstation can also receive the latest file updates through the add-on. This allows the engineer to locate the affected component directly within the BIM model. Once the correction is completed and the model is re-exported, the updated IFC

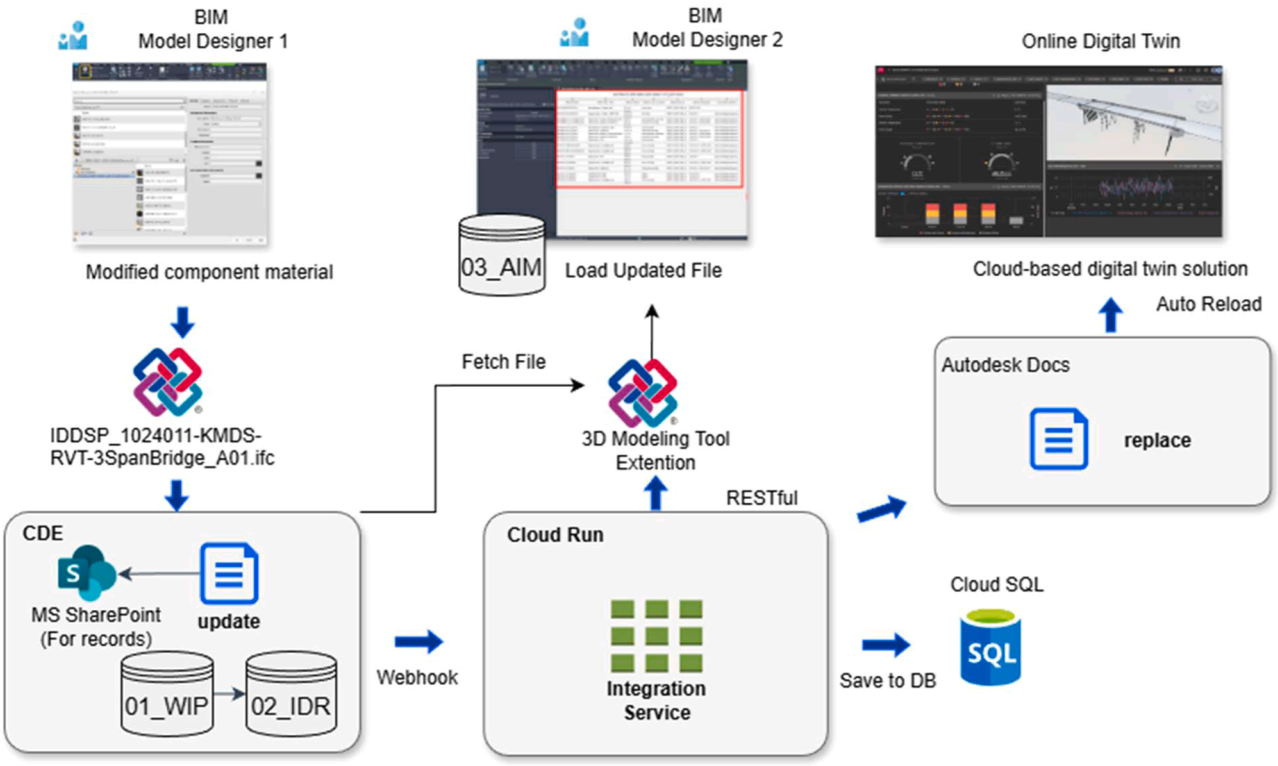


Fig. 5. The process of BIM model revision.

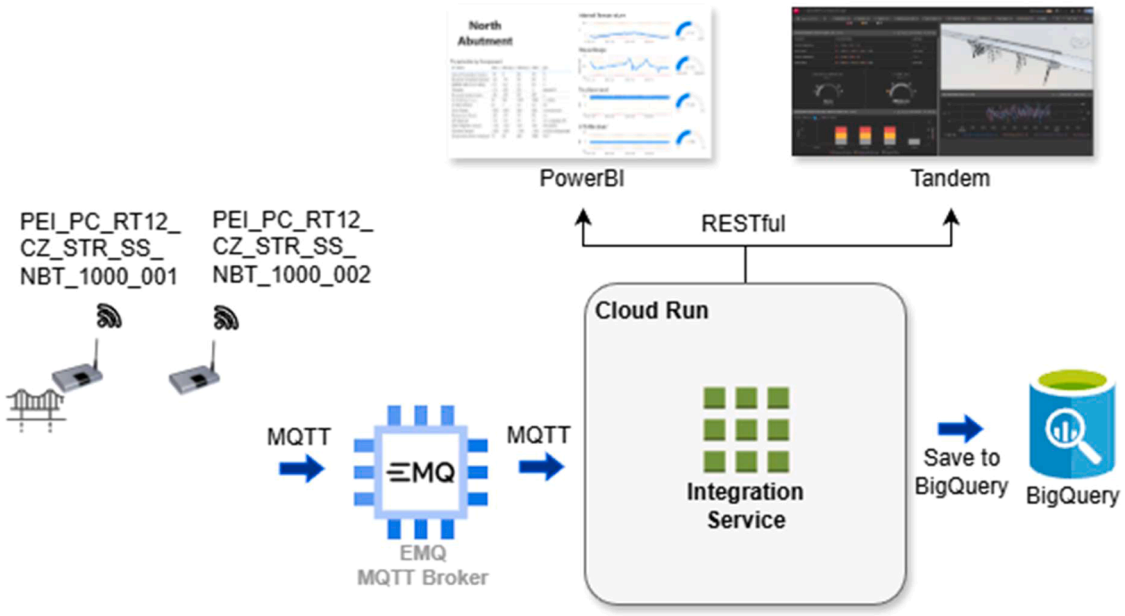


Fig. 6. Process of updating sensor data.

version re-enters the loop described in Section 4.1.1.

This mechanism ensures a seamless feedback cycle between inspection findings (IFC) and design revisions, closing the Bridge Digital Twin Information Loop.

Through these four workflows, the Integration Service transforms manual, file-based data exchange into an automated, event-driven process. Design changes, sensor updates, inspection reports, and issue management are handled within the automated information loop, driven by consistent PCIDs, unified semantics, and standardized data exchange protocols.

Beyond individual workflow validation, to test the robustness of the Persistent Component ID (PCID) mechanism, a total of 1067 component records were tracked across four data domains: BlenderBIM, IFC, Tandem, and Power BI. Randomized update tests, such as renaming elements, regenerating IFC exports, and re-uploading inspection files, were performed to examine whether identity consistency was preserved. Results showed 100% retention of PCIDs across all systems, confirming that the proposed identifier schema effectively prevents data

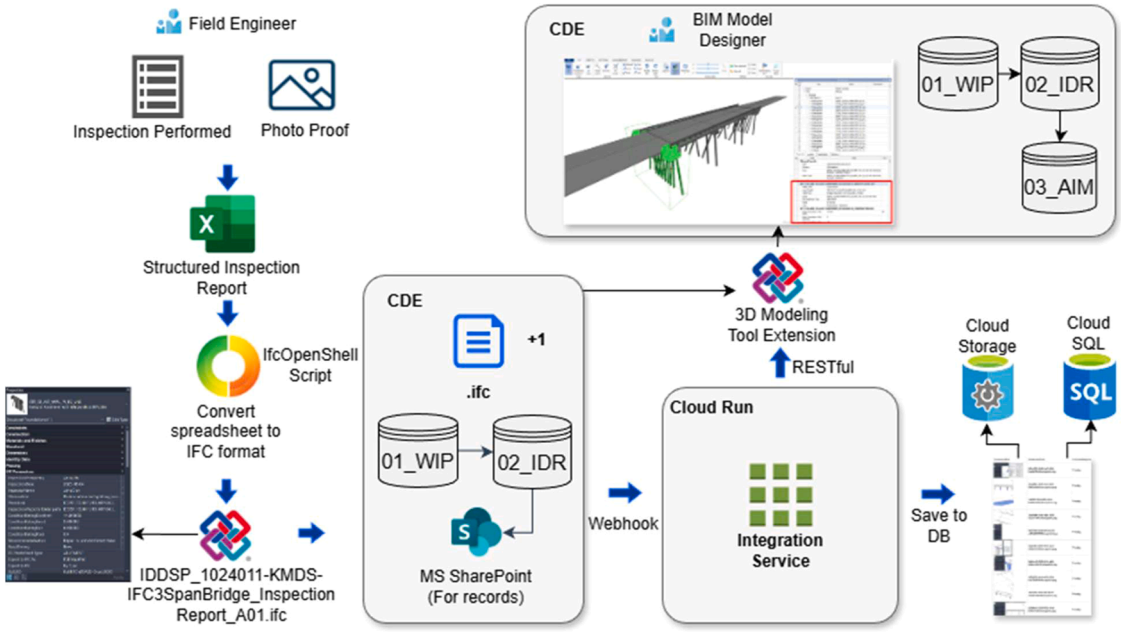


Fig. 7. Process of IFC inspection.

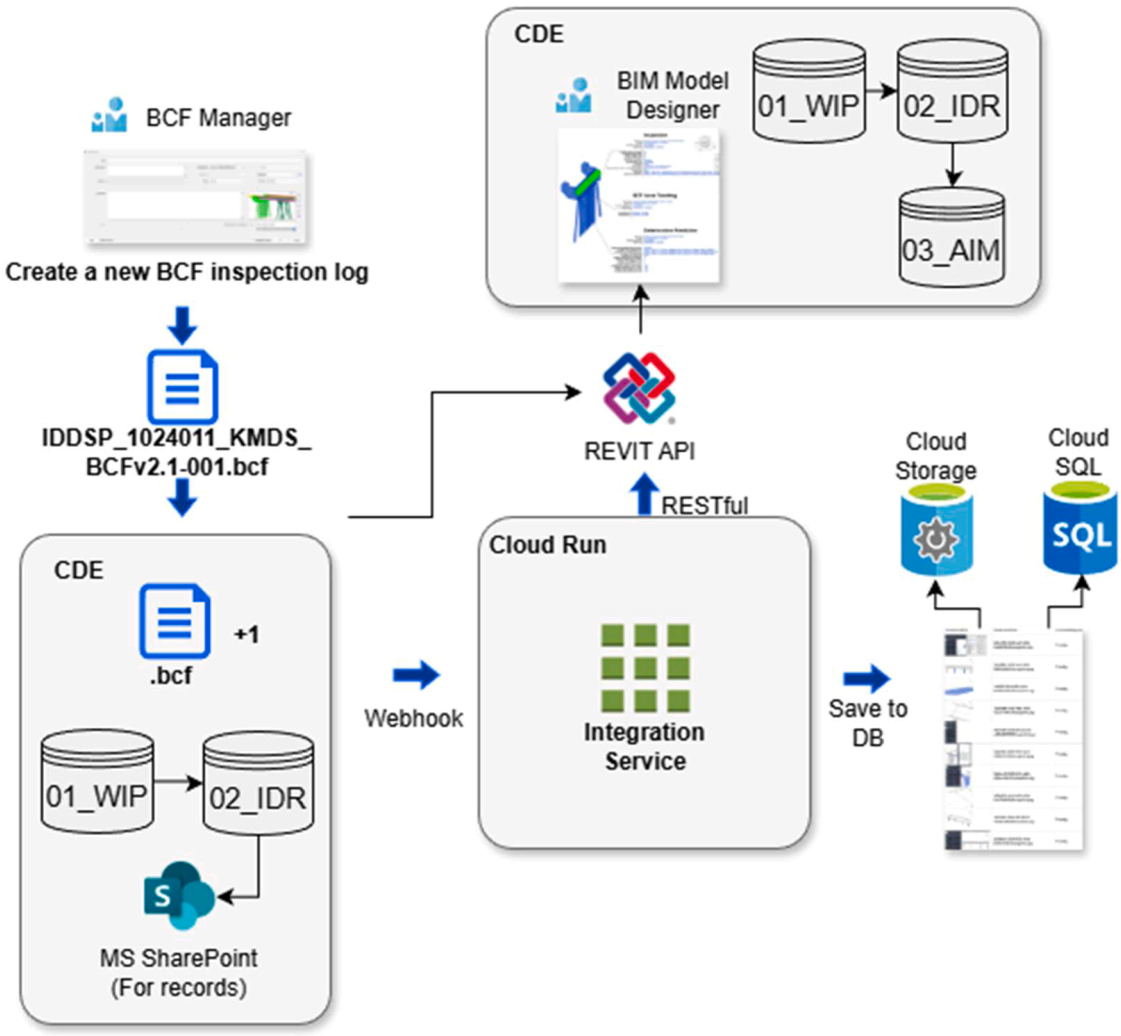


Fig. 8. Process of BCF inspection.

fragmentation and loss during multi-platform exchanges. This experiment highlights the importance of identity persistence as the foundation for semantic alignment and automated synchronization.

The "strain monitoring" scenario was extended into an automated alert case. When the strain gauge installed on Girder #001 (PCID: PEI-PC-RT12-BR_051-SP1-STR-SP-GIR-NBT_1000-PC-001) reported a value exceeding 60 με, the Integration Service triggered an event classified as OnSensorThreshold. Within 4.2 s, a visual alert appeared on both the Tandem dashboard and the Power BI analytics panel. Simultaneously, an email notification was sent to the maintenance engineer with the associated PCID and location coordinates embedded in the message body. This event-driven feedback confirmed the effectiveness of the Exchange Protocol in achieving near-real-time responsiveness.

The evaluation results corresponding to the validation criteria defined in Table 3 are assessed through the following integration scenarios. Each scenario is used to systematically verify the framework's performance in terms of identifier persistence, semantic consistency, synchronization latency, data completeness, and workflow automation.

To further evaluate the effectiveness of the proposed framework, the digital twin implementation was examined through controlled integration scenarios aligned with the validation criteria defined in Table 3. Property synchronization between the BIM model, IFC representation, and the digital twin platform was tested by modifying component attributes in the BIM environment and tracking their propagation across systems. Validation rules embedded within the semantic mapping template ensured that identifiers and property values complied with predefined constraints. The results demonstrate strong performance across all validation dimensions: identifier persistence was maintained at 100% across BIM, IFC, and digital twin platforms; semantic consistency was achieved with no observed discrepancies in mapped properties; synchronization latency remained within near real-time thresholds; and data completeness was preserved without missing values. Furthermore, the automated integration pipeline eliminated manual data handling steps, confirming improved workflow efficiency compared to conventional BIM-based processes. The case study implementation validates the practicality of the identity–semantic–interaction framework as a foundation for standardized digital twin construction. The persistent identity mechanism enabled traceable data lineage across design, inspection, and monitoring stages. The semantic protocol established a shared vocabulary that eliminated the ambiguity common in heterogeneous data environments. The exchange protocol proved scalable and tool-agnostic, confirming that open standards such as IFC and MQTT can sustain real-time interoperability in infrastructure applications. The Integration Service acted as the operational backbone of this framework, coordinating event-driven synchronization between modeling tools, IoT platforms, and analytics environments, thereby ensuring continuous and automated data flow throughout the bridge's digital lifecycle. The results demonstrated that the framework successfully overcame traditional "geometry–data disconnects" and manual handover inefficiencies.

5. Discussion

The proposed framework demonstrates a practical and modular approach to realizing a bridge digital twin within existing software ecosystems. By integrating open standards such as IFC and standardized identifiers like PCID and semantic protocol, the framework ensures data consistency and traceability across multi-source domains, including geometry modeling, inspection documentation, and sensor monitoring. This modular structure allows each subsystem to evolve independently while maintaining interoperability through the shared protocols. This method provides a clear methodology for bridging the traditional gap between design-centric BIM environments and data-centric digital twin systems. In practice, this enables the creation of "living" bridge models that not only visualize geometry but also represent real-world behaviors and conditions.

The case study results demonstrate several improvements over conventional BIM-to-IoT integration workflows. Traditional digital twin implementations often rely on manual data alignment or proprietary integration layers, which can introduce inconsistencies when models are exported or updated. In contrast, the proposed framework maintains identity persistence through the Persistent Component Identifier and ensures semantic consistency through structured property mapping templates. During the evaluation scenarios, component identifiers remained unchanged across IFC exports and platform synchronization processes, while mapped attributes were propagated automatically between the BIM model and the digital twin environment. These results indicate that the framework reduces manual data reconciliation efforts while improving traceability and interoperability across the bridge asset lifecycle.

The evaluation metrics embedded in the semantic protocol also provide a systematic mechanism for verifying integration performance. In the case study implementation, synchronization tests showed that mapped properties were consistently transferred across systems while maintaining identifier continuity. These observations indicate that the framework can support reliable data exchange in digital twin environments where multiple software platforms and data sources interact throughout the infrastructure lifecycle. From an engineering standpoint, the framework proves especially suitable for small- and medium-scale bridge projects, where resources, expertise, and budgets often limit the deployment of full-featured commercial digital twin platforms. By leveraging commonly available tools such as 3D modeling tool, Power BI, and open APIs, it minimizes the need for proprietary infrastructure while still achieving real-time interoperability. The proposed Integration Service plays a central role in this process. It automates data synchronization between design, monitoring, and decision layers through event-driven rules defined in the Exchange Protocol. This mechanism effectively transforms fragmented, file-based workflows into a continuous data ecosystem where updates flow maintain consistency between the physical and digital domains.

However, several limitations remain evident. In large-scale or multi-bridge management scenarios, the current architecture may struggle with scalability and distributed coordination. The rule-based mapping between geometric and semantic layers still relies on manual configuration of JSON templates, which increases maintenance overhead and limits real-time adaptability. Moreover, the system lacks a robust transaction layer for managing concurrent updates across teams, which could lead to synchronization conflicts in collaborative environments. The semantic protocol, while effective for schema alignment, does not yet support reasoning or ontology-based inference, restricting its use in advanced analytics and AI-driven decision-making.

When compared to commercial ecosystems such as Autodesk Tandem or Bentley iTwin, the proposed framework prioritizes transparency and interoperability over built-in intelligence and visualization capabilities. Tandem and iTwin provide integrated dashboards, performance simulation, and predictive analytics, whereas the present approach focuses on interfacing. i.e., the ability to connect heterogeneous data sources through open protocols. This distinction underscores the framework's role not as a competing platform, but as a middleware layer that can complement existing systems by providing standardized, verifiable data pipelines. In this sense, the framework can be viewed as a stepping stone toward a more federated digital twin infrastructure, where local BIM models, IoT systems, and inspection tools remain autonomous but interoperable through standardized semantics and identifiers.

Bridge owners face dual challenges of technical expertise and resource allocation when deploying digital twin platforms. Implementation requires specialized knowledge in IFC schema architecture, API integration, and template configuration, alongside the time and financial investments associated with PCID standardization and long-term system maintenance. To lower these adoption barriers, we propose an incremental deployment strategy. By prioritizing core tasks such

as integrating inspection metadata and utilizing open-source tools like IfcOpenShell alongside reusable template libraries, owners can effectively mitigate initial development costs and operational disruption, thereby establishing a low-risk pathway toward maximizing the long-term value of digital twins.

Despite these limitations, the framework contributes to the broader digital twin discourse in three significant ways. First, it operationalizes theoretical concepts, such as identity persistence and schema homology, into a practical, implementable workflow tested on a real bridge project. Second, it demonstrates the feasibility of event-driven synchronization using lightweight APIs and open formats, proving that continuous updates do not necessarily require high-cost enterprise platforms. Third, it establishes a replicable pattern for digital twin development that emphasizes openness, extensibility, and incremental adoption, which are critical attributes for public-sector infrastructure projects constrained by budget and interoperability mandates. The validation results confirm that the proposed framework not only functions as intended but also provides measurable improvements over conventional BIM-based data exchange workflows, particularly in terms of automation, data consistency, and interoperability.

By simplifying core technical workflows, this framework significantly lowers the barrier for digital twin implementation in bridge engineering. It eliminates manual IFC data loops by using the PCID to maintain a stable identity link between BIM models and IoT datasets, while reusable semantic templates replace repetitive, project-specific configurations. These improvements directly benefit key stakeholders: designers no longer need to manually cross-reference inspection records; inspectors can link monitoring data directly to 3D components; and asset managers gain immediate access to automated updates for informed decision-making via dashboards. Through this lean approach, the framework enables small-scale engineering teams to deploy robust digital twin solutions without relying on expensive enterprise platforms.

Future research should address several directions. Enhancing scalability through microservice-based architecture and message brokers would enable distributed coordination across bridges and teams. Strengthening data governance, including versioning, access control, and audit trails, will be essential for long-term lifecycle management. Furthermore, integrating rule engines could elevate the framework from syntactic interoperability toward cognitive digital twins capable of adaptive learning and predictive maintenance.

6. Conclusion and future works

This paper developed and validated a bridge digital twin framework based on an identity–semantic–interaction architecture. The framework integrates four key mechanisms; Persistent Component Identifier, Semantic Protocol, Exchange Protocol, and Integration Service, to enable consistent and interoperable data flow across bridge design, inspection, and monitoring stages. The validation was conducted through a proof-of-concept case study and controlled integration scenarios, assessing performance in terms of identified persistence, semantic consistency, synchronization latency, data completeness and workflow automation. By establishing standardized identifiers, unified property schemas, and event-driven synchronization, the system connects BIM geometry, inspection records, and IoT monitoring data within a coherent digital environment.

The pilot implementation on a representative bridge project verified the practical feasibility and performance of the proposed framework. The PCID mechanism ensured persistent identity alignment from modeling to field inspection and Tandem visualization, enabling direct traceability of each physical component. The semantic protocol formalized the attribute mapping between the model, IFC 4.3, and analytical tools through JSON templates, achieving semantic consistency and reducing data translation errors. Across the validation scenarios, the system demonstrated strong results against all defined evaluation criteria. Identifier persistence was maintained at 100%, ensuring stable component identity across BIM, IFC, and digital twin platforms. Semantic consistency was achieved through structured mapping templates, with no discrepancies observed in property values across systems. The exchange protocol and integration service automated real-time updates between heterogeneous systems, including Power BI dashboards, Tandem digital models, and IoT sensors, transforming the traditionally fragmented, file-based workflows into a continuous, event-driven data loop. This integration approach demonstrated that a dynamic, maintainable digital twin can be built using existing, widely available tools rather than fully customized software platforms. These results demonstrate that the proposed framework provides a reliable and scalable solution for interoperable digital twin implementation in bridge management.

From a management perspective, the proposed approach offers a low-cost and replicable pathway for bridge owners and operators to transition from static BIM documentation to dynamic, data-driven asset management. It allows incremental implementation: small-scale deployments can start with inspection data integration, while more advanced setups can extend to real-time IoT synchronization. This flexibility makes the framework particularly suited to public-sector and medium-scale bridge projects, where resource constraints and software compatibility often limit digital adoption.

Nevertheless, several limitations emerged during the implementation. First, the framework's current design provides limited scalability for large, multi-organizational projects involving multiple contractors, inspection agencies, and asset owners. The rule-based mapping and event handling mechanisms still rely on manual configuration, which restricts automation and may introduce maintenance overhead when managing large data volumes. Second, while the system achieves interoperability at the data exchange level, it lacks semantic reasoning and analytical intelligence, which are features necessary for advanced applications such as predictive maintenance, deterioration modeling, and performance simulation. Third, the integration service currently operates as a centralized middleware, which may become a bottleneck under distributed or cloud-based deployment scenarios.

To address these challenges, future research will explore three primary directions. The first one is cloud-based microservice architecture by replacing the centralized integration service with distributed event brokers (e.g., Kafka, MQTT) and microservice endpoints will improve scalability, fault tolerance, and parallel data handling for multi-bridge or multi-stakeholder systems. The second one is AI-assisted semantic expansion, which includes machine learning and ontology-based reasoning to enable bridge health monitoring and forecasting. The third aspect is lifecycle intelligence and feedback. The next iteration of the framework will focus on closing the data loop between structural monitoring and decision-making, to enable predictive analytics and maintenance recommendations through hybrid physical–digital simulation models. Future research will extend the validation of the proposed framework beyond the current proof-of-concept implementation. This includes applying the framework to larger bridge inventories, conducting comparative benchmarking against existing digital twin workflows, and engaging infrastructure owners and domain experts to support external validation. Additional work will also incorporate condition assessment modules and predictive deterioration models, enabling further evaluation of the framework in large-scale and operational asset management scenarios.

In summary, the proposed bridge digital twin framework represents a practical engineering pathway for realizing digital transformation in infrastructure management. By balancing accessibility, interoperability, and data richness, it bridges the gap between theoretical digital twin concepts and field-level implementation. The framework's open, extensible structure provides not only a foundation for ongoing research but also a deployable blueprint for transportation agencies seeking to modernize bridge asset management through digital twin technologies.

CRediT authorship contribution statement

M.A. Rico Thirion: Writing – review & editing, Visualization, Investigation, Formal analysis. **S. Moghtadernejad:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Y. Xu:** Writing – original draft, Investigation. **Z. Lounis:** Writing – review & editing, Conceptualization. **Q. Chen:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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